

- 1 (a)** Find the Maclaurin series for $\sin^{-1}x$ up to and including the term in x^3 . [5]

This image shows a full page of a document template designed for handwritten notes or essays. It features approximately 28 evenly spaced, thin grey horizontal lines across the entire width of the page. The margins are consistent on all sides, providing ample space for writing. There are no pre-printed questions, headings, or other markings on the page.

- (b) Deduce an approximation to $\int_0^{\frac{1}{5}} \frac{1}{\sqrt{1-u^2}} du$, giving your answer as a fraction. [1]
